GLENCORE

Data Standard
Measuring Point

12

Document properties

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1. Introduction

1.1 Master Data Standards - Purpose

The suite of Master Data Standards provide clear and prescriptive definitions and rules that support a consistent and common definition of data entities across all business systems.

It describes:

- What rules / guidelines are applied to determine how many of the relevant data entity will be created.
- What rules will be applied to determine the codes used to create the relevant data entity (e.g. Plant code or Material code);
- What rules will be applied to determine the name / short description of the relevant data entity (e.g. Plant name or Vendor name);
- Rules for the use of views / sub-entities within the Data Entity;
- Identification of the key Attributes on the data entity whose usage needs to be standardised;
- When multiple values are available for the nominated 'key' attributes, the rules definition should list the criteria to use to determine the appropriate value; and
- If applicable, any rules / guidelines for the association of the relevant data entity with other data entities.

The Data Standard is **NOT** used to document:

- 'Technical' documentation on how the system has been configured and the design rationale for the configuration
- Identifying the operational need on when these Standards should be applied, They are to provide the guidance on how the Standard is applied
- List of actual values for the data entity / attributes within the data entity except where it is required to explain the application of the data standard
- User training or procedural documentation for master data maintenance
- Data migration

1.2 Measuring Point Standard – Purpose

The purpose of this Standard is to:

- Identify the minimum Master Data field requirements for SAP Maintenance Plans and Items at Glencore.
- Ensure alignment with the Solution Building Block SBB_D90 Maintenance Master Data and Asset Management Framework Element 5

- Identify the minimum requirements for creating, maintaining, and retiring the master data
- Provide the guiding principles to manage Measurement Point master data
- Identify the primary audience for the application of this standard

1.3 Using this document

The Data Standard is targeted at the following audience:

- Master data maintenance teams in order to understand how the object needs to be maintained.
- End users supporting the maintenance function whereby appropriate knowledge is required to use the system effectively which include but are not limited to; Planners, Schedulers, Reliability Engineers etc.
- Business Intelligence team to assist with the design of reporting solutions.
- Project teams whose scope impacts the data object to ensure the solution design is compliant with the approved data standard.
- Data migration team for the build of transformation and load tools.
- Transition to support teams to assist in the development of user guides.

1.4 Governance

Master Data Governance (MDG) is the governance tool used to create, maintain and retire Master Data objects in Plant Maintenance. The Asset Management group provides the governance for all requirements described in this Standard as well as approval of any change to related Master Data.

This Standard shall be reviewed as often as the business requires but at a minimum of every five years.

The Standard is subject to Glencore's Continuous Improvement (CI) processes based on Glencore's shared experiences and Lessons Learned.

Revisions, reviews, and updates to the Standard take place under the guidance and approval of Regional Asset Management and Subject Matter Expert (SME) leaders

2. Data Object Description

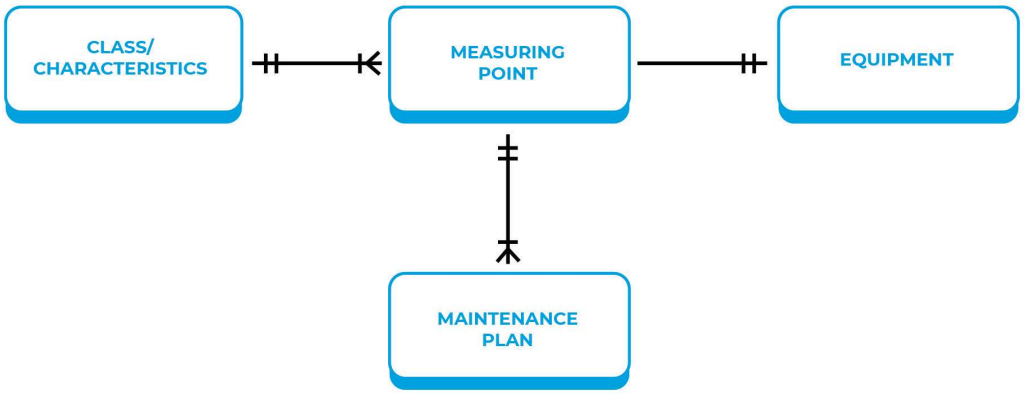
2.1 Definition

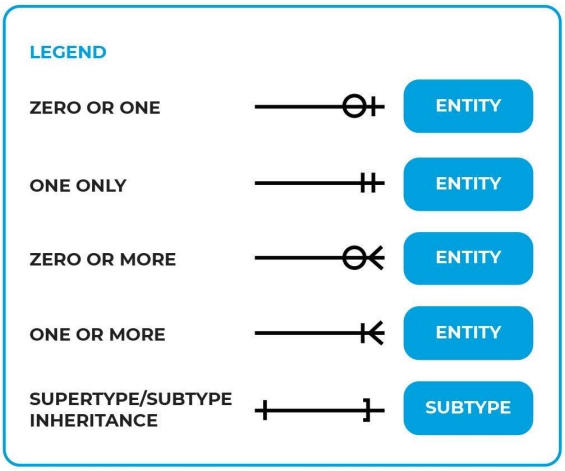
Measuring Points are created against Equipment to:

- Record physical asset utilisation – such as hours run, kilometres travelled, tonnes moved etc.
- Record the condition of a physical asset at a given time and date – such as temperature, pressure, flow rate, etc.
- Record Event History of a physical asset – such as change out dates of assets not utilising counter-based measuring points.

Actual measurements and data to support the Measuring Point are recorded on Measurement Documents. These Measurement Documents can be added manually or automatically transmitted via "In Plant" systems.

2.2 Entity Relationship Diagram (ERD)





3. Business Requirements

3.1 Business design Rules & Criteria

The design will only have one Measuring Point Category being 'M'. Measurement Points will have 2 distinct types: Counter and Non-Counter.

3.1.1 Counter Measuring Points

These Measuring Points are used to support scheduling performance-based strategy Maintenance Plans.

For Counter Type Measuring Points the annual estimate field is mandatory. This entry sets the base figure that SAP uses to calculate the schedule planned date of a Maintenance Call (Work Order) in conjunction with the entered Measurement Counter Readings. The Annual Estimate is a positive whole number.

3.1.2 Non-Counter Measuring Points (Condition Monitoring)

These Measuring Points are used to record values for condition monitoring with the potential creation of Notifications for out-of-range measurement readings. (Future configuration)

For these Measuring Points setting the Limits is mandatory; allowable Limit settings are:

- Range (Set both Lower and Upper limits)
- Lower limit only
- Upper limit only

4. Detail Standard

4.1 Data Object Coding Standard

Measuring Points will be a system generated number.

4.2 Attributes

4.2.1 Measuring Point Object

Application Definition	Measuring Point object refers to the type of SAP object, for which data is being collected
Guiding Principles	At Glencore, Equipment will be the only object for which data will be collected
Application Coding Convention	3 Characters alphanumeric
Naming Convention	The Object codes used will be the generic SAP code of: IEQ = Equipment

4.2.2 Measurement Position

Application Definition	A description with which you can describe the position or type of reading of the measuring point.
Guiding Principles	Counter based Measurement Points will typically be derived from the Characteristic. e.g. Characteristic PM_HOURS – Measurement Position HOURMETER. For Non-Counter based Measurement Points it will be based on what is being measured. i.e. Carbon Dioxide (Idle) – EA_CO2ID
Application Coding Convention	20 Characters alphanumeric
Naming Convention	Free Text

4.2.3 Measuring Point Characteristic

Application Definition	Measuring Point Characteristic defines a single custom defined field which is used to further define the properties, attributes or values of the measurements that will be captured for the Measuring Point.
Guiding Principles	The Characteristics that will be used for Measuring Points are listed in Appendix A 5.1.
Application Coding Convention	30 Characters Alphanumeric
Naming Convention	As per Appendix A

5. Appendix A: Additional Information

5.1 Measuring Point Characteristics

The following Characteristics will be used for Measuring Points

Characteristic	Characteristic Description
PM_ALARM	Alarm
PM_CHARGE	Charge
PM_CYCLE	Cycle
PM_DRILL_METERS	Drill Meters
PM_EMISSIONS_PPM	Emissions Analysis
PM_HOURS	Hours
PM_KILOMETERS	Kilometers
PM_KILOWATT HOURS	Kilowatt Hours
PM_LITRES	Litre
PM_MILLIMETERS	Millimeters
PM_PARTICULATES_MGQ	Particulates mg/m3
PM_PERCENTAGE	Percentage
PM_PRESSURE	Pressure
PM_TEMPERATURE	Temperature
PM_TONNES	Tonnes
PM_VALUE	Value

5.2 Complete Field List

- **Table –** SAP table where data object appears
- **Field –** SAP data attribute field
- **Field Name –** SAP field name of data object
- **Description –** Description of input required
- **Data Type –** SAP data type (CHAR=Character, NUMC=Number)
- **Length –** Character length of field
- **MCI –** Data requirement (M=Mandatory, C=Conditionally mandatory, I=Internally populated)
- **Field Validation Type –** Entry type allowed (e.g. Free text, List selection) and additional information

Table	Field	Field Name	Description	Data Type	Len	M/C/I	Field Validation Type
RIMRO	MPOTY	Measuring Point Object	The key for the category of the object at which the measurement point is located	CHAR	3	M	IEQ for Equipment
EQUI	EQUNR	Equipment	Number by which a piece of equipment can be clearly identified	CHAR	18	M	Valid Equipment number
IMPT	MPTYP	Measuring Point Category	Key used to identify a measurement point category	CHAR	1	M	M – Measurement Point General

Data Standard - Measuring Point

Table	Field	Field Name	Description	Data Type	Len	M/C/I	Field Validation Type
IMPT	INDCT	Measuring Point Counter Indicator	Indicator showing that the measuring point is a counter	CHAR	1	C	Select for Counter based Measurement Points
IMPT	POINT	Measuring Point	Key identifying the measuring point	CHAR	12	I	A unique system generated number.
IMPT	PSORT	Measuring Position	A description to describe the position or type of reading of the measurement point	CHAR	20	M	Measuring Position at which the reading is taken. See Measurement Position
IMPT	PTTXT	Description of Measuring Point	Description of the measuring point in the form of short text	CHAR	40	M	Measuring Point short Text Refer to Data Standard - General Configuration
IMPT	ATNAM	Characteristic	Name that uniquely identifies a characteristic	CHAR	30	M	Appendix Measuring Point Characteristics contains valid entries.
IMPT	DECIM	Decimal Places	The number of decimal places with which numerical values for the characteristic are to be represented	NUMC	5	M	If a reading is required that includes decimal places to a certain degree of detail – then update this field with the necessary decimal places – otherwise the default is two decimal places.
IMPT	TRANS	Measuring Point from which Measuring Reading was Transferred	Measuring point for which the counter reading is entered and from which counter reading differences are copied	CHAR	12	C	Mandatory for Measuring Points that inherit readings.

Table	Field	Field Name	Description	Data Type	Len	M/C/I	Field Validation Type
IMPT	INDTR	Indicator that Measurement Reading Transfer is Supported	Indicator that measurement reading transfer is desirable for this measurement point.	CHAR	1	C	Always checked when "Transfer of" is populated. Not to be checked when "Transfer of" is not populated
RIMRO	PYEAC	Annual Estimate Reading	Annual estimate for the object to which the counter applies. This field is used in conjunction with maintenance plans to calculate the necessary maintenance intervals	NUMC	22	C	For counter-based Measurement Points only. Numerical value
RIMRO	MRMAC	Upper Range Limit	The highest measurement reading that can be entered in a measurement document for this measurement point to trigger an out of tolerance event	CHAR	22	C	Mandatory for Non counter based Measuring Points if an Upper limit is required to trigger Notification creation.
RIMRO	MRMIC	Lower Range Limit	The lowest measurement reading that can be entered in a measurement document for this measurement point to trigger an out of tolerance event	CHAR	22	C	Mandatory for Non counter based Measuring Points if a Lower limit is required to trigger Notification creation.

6. Stakeholder Signoff

Stakeholder Name	Role	Contact	Stakeholder Sign-off
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